

★ MOVIE RATING PREDICTION WITH PYTHON

☀ CODSOFT DATA SCIENCE INTERNSHIP – TASK 2

OBJECTIVE

The objective of this project is to develop a Machine Learning model that predicts movie ratings based on features such as genre, director, and actors. This project helps in understanding regression algorithms, data preprocessing, and predictive analytics.

PROJECT DESCRIPTION

The Movie Rating Prediction project includes:

-  Movie dataset loading and analysis
-  Data cleaning and preprocessing
-  Converting categorical data into numerical format
-  Training a Machine Learning regression model
-  Predicting movie ratings
-  Evaluating model performance

TOOLS USED

 Python

 VS Code

 Pandas

 Scikit-learn

 NumPy

FEATURES

1. Dataset Loading

Loads movie data from CSV files.

2. Data Cleaning

Removes missing values and prepares the dataset.

3. Data Preprocessing

Converts genre, director, and actor names into numerical values using Label Encoding.

4. Model Training

Uses Random Forest Regressor to train the prediction model.

5. Rating Prediction

Predicts movie ratings based on movie features.

6. Performance Evaluation

Measures model performance using Mean Absolute Error (MAE).

CONCEPTS USED

- ✓ Data Cleaning
- ✓ Data Preprocessing
- ✓ Label Encoding
- ✓ Machine Learning
- ✓ Regression
- ✓ Random Forest Regressor
- ✓ Mean Absolute Error (MAE)

KEY LEARNINGS

- ✓ Understanding regression models
- ✓ Working with real-world movie datasets
- ✓ Converting categorical data into numerical format
- ✓ Training and evaluating Machine Learning models
- ✓ Improving data analysis skills

INSIGHTS

Machine Learning can estimate movie ratings using available movie information.

Data preprocessing plays an important role in prediction accuracy.

Random Forest Regressor is useful for regression-based prediction tasks.

Quality and quantity of data influence model performance.

CONCLUSION

The Movie Rating Prediction project successfully demonstrates how Machine Learning can be used to predict movie ratings based on genre, director, and actor information. This project enhanced my understanding of regression algorithms, data preprocessing, and predictive modeling as part of the CodSoft Data Science Internship.

OUTPUT

```
Movie Rating Prediction Completed Successfully!  
Mean Absolute Error: 1.03
```

```
Sample Predictions:
```

```
Actual Rating: 7.4 | Predicted Rating: 6.0  
Actual Rating: 4.9 | Predicted Rating: 5.7  
Actual Rating: 6.5 | Predicted Rating: 6.4  
Actual Rating: 5.7 | Predicted Rating: 6.5  
Actual Rating: 7.0 | Predicted Rating: 5.7
```

✓ Movie Dataset Loaded Successfully

✓ Data Cleaning Completed

✓ Label Encoding Applied

✓ Random Forest Regressor Model Trained

✓ Movie Ratings Predicted Successfully

✓ Mean Absolute Error (MAE): 1.03



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